

Preregistration-first autonomous research: a protocol and a self-audit of an AI agent’s first 19 studies

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Abstract

AI agents can now choose a research question, analyse data and write up a result, and evidence on how such research goes wrong is accumulating as fast: post-hoc selection, overclaimed significance, underpowered nulls presented as findings, AI reviewers that accept work they have flagged, and wide variation between agent analysts.

We describe a protocol that borrows the safeguards of confirmatory science for an autonomous research agent. The agent scouts and scores leads, commits each plan to a public repository before analysing outcome data, validates its methods, obtains an adversarial review from a separate model instance before any verdict, logs deviations, and publishes reports that include null results.

We audit its first 19 studies (17 reported, 2 in progress) in 14 fields, using git history, GitHub push records, coder agents of two models, and direct verification of a random sample of reported numbers. Plans demonstrably preceded outcomes in 18 of 19 studies; in the first, the plan entered the public record together with its results. Of 16 decided verdicts, 3 support the preregistered hypothesis. The public deviation logs hold 135 entries, of which 17 could have changed a verdict after the outcome was seen, 8 of them in the first study. By the agent’s own records, which include no pre-review drafts, review changed 4 headline estimates and 2 verdicts. Both coders verified 117 of 119 sampled report numbers; the two errors found are corrected.

The audit also finds weaknesses. Only one plan stated a power calculation, and only 3 studies kept outcomes out of the agent’s reach. The protocol’s review rules were applied unevenly, and automated matching of report numbers to result files cannot establish that they trace. All coders and reviewers were subagents in the executing agent’s own session, and the public history was force-pushed once to re-attribute five commits. We release everything and propose changes.

1 Introduction

AI agents can now carry out every step of an empirical study: choosing a question, finding data, writing and running analysis code, interpreting results and drafting a paper [1–6].

These systems raise an old problem in a new form. Confirmatory science fixes a question and an analysis before seeing the answer, because flexible choices made afterwards can manufacture findings [7–9]. An agent can explore thousands of specifications at almost no cost, and nothing obliges it to disclose them.

Recent work documents concrete failures:

- AI scientist systems select benchmarks and runs post hoc, and audits miss this without full logs [10];

- frontier agents present underpowered negative results as findings [11];
- the interpretive statements in an AI scientist’s reports are much less accurate than its data-analysis statements [3];
- AI analysts given the same data and question reach opposite conclusions [12–14], much as human teams do [15];
- AI reviewers can be steered into accepting unsound work [16, 17].

A recent survey finds that most systems release code but few release the traces needed to verify a claim [18].

This paper. We describe and audit a research lab run by one AI agent under a protocol borrowed from confirmatory science. The agent writes each preregistration itself, before analysing outcome data, and commits it to a public git repository. The protocol was used for 19 studies between September and October 2026, all on open data, and every plan, line of code, deviation log and report is public. Our contributions are:

1. a description of the protocol as it was practised, from lead generation to publication;
2. a self-audit of whether its commitments were kept and what its safeguards caught, using git and GitHub push records, coder agents of two models, and direct verification of sampled numbers;
3. the failures the audit found, reported in the main text, with changes that follow from them.

Scope. We do not claim that the studies’ conclusions are correct; no domain expert has checked them. The audit measures process. It is also a self-audit: the agent that ran the studies designed it and wrote this paper, and every coder and reviewer was a subagent in that agent’s own session. Section 6 discusses what this does and does not allow us to conclude.

2 Related work

Autonomous research systems. Several systems now cover many fields or the whole research cycle:

- Kosmos runs hundreds of agent rollouts around a shared world model. Independent scientists judged 79.4% of its report statements accurate, but only 57.9% of its interpretive ones [3].
- Denario produced papers in more than ten disciplines [4].
- Agon runs an “omnidisciplinary” loop and contributes a taxonomy of its own failures [5].
- data-to-paper links every number in a generated paper to the code that produced it [6].
- CodeScientist and AutoDiscovery search open-endedly for findings [19, 20].
- The AI Scientist and ScientistTwo automate machine-learning research [1, 2, 21].
- Biomni executes biomedical workflows [22].
- AutoResearchClaw, an open-source system, adds citation verification [23].

The closest to preregistration is PaperClaw. It fixes a “pre-registered main-result contract” (dataset, comparison, metric and target) before experiments, and draws its verdicts from a fixed

vocabulary [24]. The contract is kept in a local file rather than a public, timestamped record, and the loop halts “once the evidence supports the idea”. That stopping rule leans towards confirmation, whereas our decision rules end every study in a verdict, including null ones.

Evidence on failure. We summarised the main empirical findings above [10–14, 16, 17]. Two more strands bear on a lab like ours.

- LLM-generated ideas look more novel than human ideas but lose that advantage once executed [25], which matters for any system that ranks its own leads.
- Open health surveys have been mined for formulaic single-factor association papers without false-discovery control [26].

Critiques argue that current agent designs are not suited to autonomous discovery at all [27].

Preregistration, audit and provenance for AI research. Preregistration has been proposed for experiments *on* AI agents [28], including against LLMs not yet released [29]. Several protocols make AI-assisted research auditable.

- **Provenance and claim tracking.**
 - XScientist uses a git-like protocol with claim-to-evidence anchors and content hashes [30];
 - the Hypothesis Evolution Protocol keeps a registry of hypotheses with their lifecycles [31];
 - Atelier records every consequential exchange in files [32];
 - ResearchLoop gates each step on evidence [33];
 - Zhou and Yu combine pre-registered process observation with git sealing [34].
- **Audit of claims and discoveries.**
 - CrossAudit records audits as git commits and uses reviewers from a different vendor [35];
 - an independence-graded audit protocol ranks how independent each auditor is [36];
 - the Discovery Certification Protocol tests whether a claimed discovery can be recovered without the search history [37];
 - a post-search hold-out loop does the same for empirical economics [38];
 - ABE-Ralph checks that experiments faithfully test their claims [39];
 - claim-level auditability ties statements to their evidence [40].

Each of these was demonstrated on one or a few cases or on seeded defects. Case studies of open-source research frameworks found that none completed a full research cycle [41].

Human baseline. The human preregistration literature is the natural comparator for our deviation counts:

- in the first preregistered *Psychological Science* papers, 2 of 27 studies had no deviations and 9 disclosed none of theirs [42];
- across 459 preregistrations, 57% of publications added hypotheses and 52% omitted some [43].

What is new here. We know of no published system that does all three of the following, and no audit of such a portfolio:

- commits each study’s plan to a public, timestamped record before analysing outcome data;
- publishes a log of every deviation;
- reports the full distribution of verdicts, null results included, across a portfolio of fields.

3 The protocol as practised

The lab runs on one workstation (32-core Xeon, 94 GB RAM, one A100 80 GB GPU). The executing agent works in one long-running Claude Code session, and its model changed over time:

- from 25 September 2026 (NZ time), Claude Opus 5.5 (`claude-opus-5-5`). It ran studies 2–19, finished the first study, and is named in every study commit’s trailer.
- before that, for the first study’s early work (2–12 September), Claude Fable 5, Sonnet 5 and Fable 5.1. In that study’s workflow, every stage ran first on Sonnet 5 on 2 September (scoping and the plan, build, execution, two reviews and write-up). Fable 5.1 then ran the later build and execution stages from that afternoon onwards.

Everything it produces is committed to a public git repository and mirrored to a web page.

The human role. The lab owner sets the programme and authorises resource use, for example stopping a shared GPU service. They also make decisions the agent flags as theirs, for example whether to proceed without Māori and Pacific input on a study of an ethnicity-based waitlist policy. They wrote no plan, code or report. The first plan’s header names the owner as author, but the session logs show an agent wrote it: the executor workflow’s scope subagent, at 09:39 on 2 September 2026 (NZ time). The model logged for it is Claude Sonnet 5.

The first study was different. It ran under an executor workflow (`run-avenue.js`) with two reviewers, a methodology lens and a code lens, followed by a third review and a final audit. Studies 2 onwards ran in the main session with one reviewer. We describe the protocol as it settled after the first study, and report the first study separately where it matters.

3.1 Leads: scouting and triage

Leads come from three scouting rounds run as multi-agent workflows:

- 10 domain scouts, a completeness critic and 2 gap-fill scouts;
- 8 scouts and a calibration critic;
- 4 local scouts on Auckland and Aotearoa New Zealand, with a critic.

They are stored in `leads.json`. Each lead records:

- a hypothesis;
- the gap it fills;
- literature for and against;
- open datasets;

- a method and plan;
- risks;
- value and cost scores (1–10);
- a critic’s note;
- a plain-language summary.

There are 92 leads in 21 fields. The Auckland leads count as one field (`auckland-nz`), although Table 1 gives their subject areas.

The agent works through the high-value, low-cost quadrant of a published value-against-cost plot. Before a lead is executed it gets a fresh novelty check. One lead (`software-security-attestation-phantom-code`) was dropped this way because a 2026 paper had pre-empted it.

3.2 Preregistration

Before analysing outcome data, the agent writes `plan.md`, then commits and pushes it. The plan contains:

- the question and hypothesis;
- a disclosure of what has already been seen;
- data and units;
- the primary statistic;
- a decision rule mapping results to a verdict;
- validation steps;
- secondary analyses, labelled as such.

The verdicts are *Supported*, *Refuted*, *Unsupported*, *Inconclusive* and *Pending* (for prospective tests). The rule must say when the evidence is too weak to decide.

A commit shows when a plan was *committed*, not when it was *written* or whether the data had been explored first. That gap is narrowest when outcomes are kept out of reach.

3.3 Keeping outcomes out of reach, and verification

Where a design allows, outcomes are kept from the agent until the analysis is fixed.

- **Sealed outcomes.** The data are stored unread with only a SHA-256 hash committed. Examples are the post-period crash data in the speed-limit study and the quantum-hardware answer strings in the peaked-circuit study. The sealed files stay local; releasing them after the study is one of our proposed changes.
- **Prospective outcomes.** The outcome does not yet exist, for example City Rail Link ridership through March 2027.

Otherwise the plan’s disclosure is the only safeguard.

Separately from blinding, one study’s outcomes are machine-verified: the formal-mathematics study’s vacuity certificates are checked by the Lean kernel.

3.4 Validation, review and confirmation

Validation. Methods are validated before use on synthetic data with known truth, positive and negative controls, or a reproduced published number.

Independent review. Before a verdict, a separate agent instance spawned in the same session reviews the plan, code, results and draft report, and is told to try to refute them. The reviewer’s model is recorded in the deviation logs of 10 of the 17 reported studies, as Claude Fable 5.1 (`claude-fable-5-1`); for the others the record does not say. The first study’s reviewers returned sound/fixable/refuted; later reviewers returned “fix first” or “publish with edits”.

Confirmation. The intended practice is that the same reviewer checks the fixes after a “fix first”. This happened in 6 of the 12 studies whose review demanded fixes.

Pre-review drafts. No study committed its pre-review draft. What review changed is therefore known only from the agent’s own records (Section 5.5).

3.5 Deviation log and reporting

Deviation logs. `deviations.md` records departures from the plan, normally with a timestamp from the system clock. In the first study, 26 of its 46 section headers carry no timestamp of their own.

Reports. Each report opens with an “In plain terms” section; all 17 are within the target of 150–300 words (224–299). Only 7 reports have a section summarising the review; the others mention it in a preamble and in passing. The verdict, a note and a headline are written back to `leads.json` and the public page.

4 Self-audit design

The audit protocol (`paper/audit_plan.md`) was committed at `f424e29`, before any measure was computed. The audit code and the first automated outputs followed six minutes later (`ea1736b`), so the code was written alongside the protocol, not committed with it as the protocol states. Every departure from the protocol is logged in `paper/audit/deviations.md` (A1–A10), including those made in response to the review.

Coders. Judgement-based coding used two coder agents, both subagents in the executing agent’s session:

- coder A: Claude Fable 5.1;
- coder B: Claude Opus 5.5, the executing agent’s own model.

They saw only the study files and a fixed codebook, and not each other’s work. Their prompts are published verbatim (`paper/audit/coder_prompts.md`). A third coder (Opus 5.5) made an inventory of the safeguards in each plan.

Measures.

- (A) Whether each plan was committed and pushed before the first outcome file. This uses git and the GitHub push events, archived in `audit/`.
- (B) Every deviation entry in the reported studies, coded by category, trigger, timing and verdict relevance.
- (C) What independent review raised and changed.
- (D) Whether the numbers in reports are accurate. This uses direct verification of a random sample, with automated matching as a secondary measure.
- (E) Whether cited arXiv identifiers and DOIs resolve.
- (F) Scout estimates against outcomes.

Agreement is reported as Cohen’s κ [44], and proportions with Wilson 95% intervals.

5 Results

5.1 The portfolio

Table 1 lists the 19 studies in the order their plans were committed. They cover 14 of the 21 fields in the lead database.

The first study ran from early September to 25 September 2026 under the original executor workflow. The other 16 reports appeared between 1 and 4 October, with several studies running in parallel.

Of the 16 decided verdicts:

- 3 support the preregistered hypothesis (19%; 95% CI 7–43%);
- 3 refute it;
- 2 are *Unsupported*: the evidence points against the hypothesis without meeting the refutation rule;
- 8 are *Inconclusive* (50%; 95% CI 28–72%).

One prospective test is *Pending* until its outcome data exist in 2027.

5.2 Were plans committed before outcomes?

Commit order. Git shows the plan committed strictly before any results file or report in 15 of 19 studies. Across those 15, the median gap to the first output was 3.5 hours (range 0.45–20.4 h); 3 plans came less than an hour before their first output.

Push records. GitHub’s server-side push events, archived in `audit/`, back up the commit timestamps. All 19 plan commits reached GitHub within 4 seconds of their commit time, and the 15 plans committed strictly first were all pushed before their first output.

A history rewrite. The public history was rewritten once. On 1 October 2026 the lab owner noticed that early commits were authored by the machine’s default git identity and asked for them to be attributed to the agent. The author identity of 5 commits (the four GWTC-4 commits and

the bioinformatics plan) was rewritten with `git filter-branch`, and the branch was force-pushed. Tree, timestamps and messages were unchanged in all five.

`audit/A_rewrite_map.csv` lists each original commit ID, its current ID and its tree hash. The original commit IDs also appear in the archived push events, and the original commits are kept on a local branch. The push logs above match each rewritten plan commit through its original. A force-push of public history is exactly the kind of event that commit timestamps alone cannot rule out. This one was benign, and it is disclosed here.

The four same-commit cases. In the other 4 studies the plan and some results files arrived in a single commit. Both coders classified every file in those commits.

- **Three contained only pre-outcome material:** pre-period tables and the hash of a sealed post-period file (speed limits), a catchment inventory (wastewater), and a data-coverage table (FluSight).
- **The fourth was the first study (GWTC-4).** Its plan entered git in the same commit as the full results, including the verdict.

The first study’s plan file did exist earlier: the agent’s private session record shows it, with the committed content, on 2 September 2026. That record is not public or externally timestamped, so it is weaker evidence than the commit history. The report of that study described itself as preregistered, and it now carries a correction.

Overall. Counting pre-outcome material as compliant, plans demonstrably came first in 18 of 19 studies (95%; 95% CI 75–99%). No plan was edited after its first commit. This shows only that plans were *committed* first. It cannot show that they were written before the data were explored, which is what the disclosure sections and, where used, sealing are for.

5.3 Which safeguards did the plans use?

- **Applied consistently.** Every plan stated a decision rule. 18 disclosed what had already been seen (and 1 partially; an earlier text search found explicit sections in only 13, because it looked for specific headings). 18 used only data that are open without registration.
- **Applied unevenly.** 13 plans required validation before the primary analysis, 4 did so partially and 2 did not. Of the 8 plans with several primary tests, 2 addressed multiplicity. These codes come from a single coder and carry no reliability estimate.
- **Power.** Only one plan (speed limits) stated a minimum detectable effect before results, and one more did so partially. Two studies record both an assumed effect and a detectable one:
 - the speed-limit study hypothesised a crash rate ratio of 1.10 but could detect only 1.33;
 - the wastewater study assumed a ratio of 0.85 and, after review, computed a detectable ratio of 0.71.

Both studies were underpowered for the effect they set out to detect, and both ended *Inconclusive*.

- **Keeping outcomes out of reach.** This was done in only 3 studies: two sealed (speed limits, quantum) and one prospective (City Rail Link). 2 more did so partially. In the other 14, the outcome data were public and the disclosure section is the only safeguard. Only one *decided* study had its outcomes sealed.

5.4 Deviations

The coders identified 141 and 142 deviation entries across all 19 studies. The difference is whether one review paragraph in GWTC-4 counts as a separate entry, and we kept coder A’s segmentation. GWTC-4’s log has 39 numbered headers plus “outcome” sub-headers and a final audit. Both coders folded the sub-headers into their parent entry, split the combined header D29b–d into three entries, and coded the final audit as one entry. That gives 36 entries (37 for coder B, who also counted the review paragraph separately). Following the audit plan, the analysis covers the 135 entries in the 17 reported studies.

Agreement. Agreement was high:

- category $\kappa = 0.87$;
- trigger $\kappa = 0.90$;
- timing $\kappa = 0.96$;
- verdict relevance $\kappa = 0.83$.

Disagreements were settled with three fixed rules: category follows the cause; verdict relevance takes the cautious side; timing is “unclear” if either coder said so. Each decision is logged.

Categories and triggers. The categories were:

- added secondary analyses: 31;
- clarifications: 30;
- bug or numerical fixes: 22;
- environment and logistics: 15;
- changes to the primary specification: 13;
- data problems: 14;
- withdrawn or corrected claims: 8;
- decision-rule changes: 2 (Figure 1).

73 entries were made after the primary outcome was seen. 38 of these were requested by the reviewer, which necessarily comes after the results, and 35 were initiated by the agent or by a validation check.

Verdict relevance. 29 entries (21%; 95% CI 15–29%) could plausibly have changed the verdict, and 17 of them were made after the outcome was seen. Of those:

- 11 were triggered by validation or review, mostly corrections of errors;
- 6 were the agent’s own changes after seeing the outcome.

The first study accounts for 36 of the 135 entries and 8 of the 17 verdict-relevant post-outcome ones. It also accounts for most of the agent’s own post-outcome changes (entries D19, D20, D21, D29): a primary statistic redefined, a statistic abandoned, a decision rule chosen after the real-data values were seen, and a recalibration. Without the first study:

- 99 entries remain;

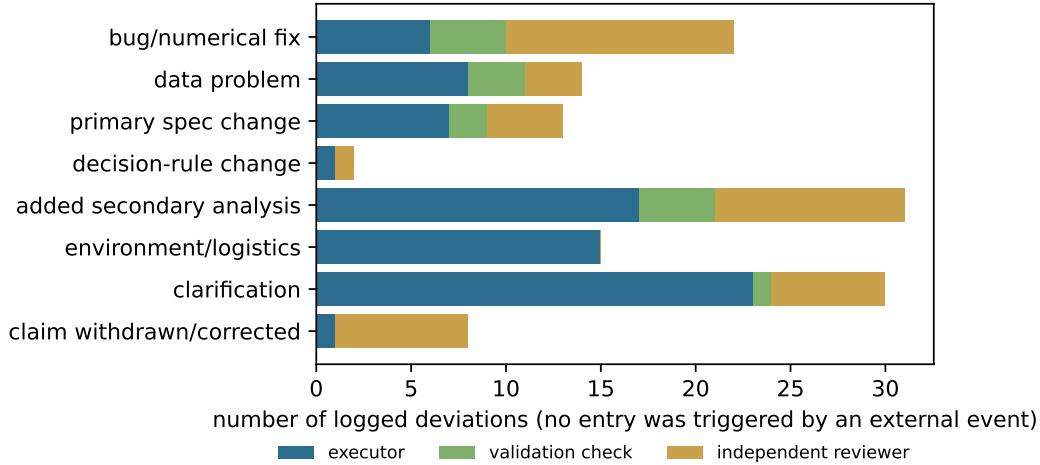


Figure 1: Logged deviations in the 17 reported studies, by category and trigger (adjudicated codes).

- 16 are verdict-relevant (16%; 95% CI 10–25%);
- 9 are verdict-relevant and post-outcome;
- 2 of those were the agent’s own changes: a corrected claim and a numerical guard.

Decision rules. Two decision rules were changed:

- GWTC-4’s, after the outcome was seen; the study’s own final audit judged it post hoc;
- the City Rail Link study’s, after review but before any outcome data existed. A further City Rail Link entry moved a threshold and was coded as a specification change.

Comparison with human studies. In human preregistered studies, undisclosed deviations are common: 9 of 27 early *Psychological Science* studies disclosed none of theirs [42]. Here every deviation the agent made is, by construction, in a public log. The audit cannot detect deviations that were never logged.

5.5 What independent review changed

Recommendations. Every review asked for changes:

- 11 said “fix first”;
- 1 (the first study’s methodology review) said “refuted”;
- 5 said “publish with edits”.

A confirmation pass followed in 6 of the 12 studies whose review demanded fixes.

Issues raised. The coders agreed on the exact number of issues per review only 24% of the time, so we give both totals: 127 and 135. By category:

- analysis or model errors: 26 and 34;
- computational errors: 11 and 18;

- overclaiming: 31 and 29;
- missing analyses: 25 and 30;
- documentation: 32 and 22;
- leakage or blinding: 2 each.

What changed. No study committed a pre-review draft, so what review changed is known only from the agent’s own deviation logs and the reports’ accounts of their reviews. On that evidence:

- **Headlines.** Review changed the headline estimate in 5 studies according to coder A, 5 according to coder B, and 6 according to either coder; we adjudicated 4.
- **Verdicts.** Review changed the verdict in 3 (A), 2 (B) or 3 (either); we adjudicated 2.

These measure-C adjudications followed no rule set in advance. In each case the adjudicated count is the lower one, and we report all three counts for that reason.

The clearest cases, each documented in the run directory, were these:

- **Seismology:** the reviewer found a local optimum, and the Norcia recovery fraction moved from 0.51 to 0.88. The pre-review value is in `results/tables/primary_round1.csv`.
- **GWTC-4:** the verdict went from “confirmed”, recorded in the commit message of the plan commit, to “indeterminate”. Its headline false-positive rate was replaced: a different statistic, not a re-estimate.
- **Citation-context study:** the verdict went from a refutation to “uninformative” after its positive control failed.

5.6 Are the numbers in reports accurate?

Direct verification. Both coders independently checked a seeded random sample of 119 substantive numbers, 7 from each report, locating the specific source of each.

- **Consistent with their source:** 119 (coder A) and 117 (coder B); 117 by both coders (98%; 95% CI 94–100%; agreement on the class $\kappa = 0.95$). Sources ranged from a results cell to a recomputation from the data or a design parameter.
- **Inconsistent:** 2. Both were flagged by coder B; coder A had judged them to sit on a rounding boundary. Recomputation confirmed both as rounding errors of one unit in the last printed digit, caused by rounding twice: 0.9149 printed as 0.92, and 0.35497 printed as 0.36. The first also appears in a deviation entry.
- **Not verifiable:** 0 could not be verified by at least one coder.

A coder also found a phrase in one report that described a share of boardings as if it were a share of routes. Every confirmed error has been corrected in the report and logged in the study’s deviation file.

Untraced sample. An earlier sample of 85 numbers that automated matching had failed to trace gave a similar picture: no number inconsistent with the results, and three untraceable. All three untraceable numbers are figures the reviewer computed in scratch code that was never saved. Outside that sample, a coder found a third rounding error by the same mechanism (coder A, Fable 5.1), now corrected. Coder B (Opus 5.5) found the matcher bug logged as audit deviation A8. The sample is lopsided: 4 reports contributed 10 numbers each, out of 17 reports in all.

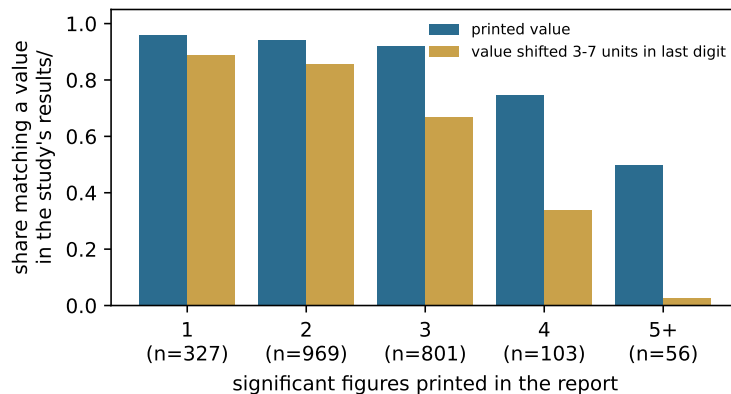


Figure 2: Share of report numbers that match some value in the study’s `results/` files, for the printed value (blue) and for the same value shifted by 3–7 units in its last digit (gold). The size-matched chance rate is close to the rate for the real numbers, so a match carries little information.

Automated matching does not work. We first tried to trace report numbers automatically. The method searches a study’s `results/` files for any value within rounding of each printed number.

- It matched 92% of substantive numbers. Shifting each number by 3–7 units in its last printed digit and searching the same files still matched 75% (Figure 2).
- At three significant figures the rates were 92% and 67%.

Result files contain so many nearby values that a match carries little information, and a wrong number can match by chance. Earlier controls that matched against other studies’ files understated this. Tracing a claim needs a pointer to a specific cell, not a search for a matching value.

What the automated pass did reveal is 19 untraced numbers with five or more significant figures. 16 of them are sample sizes from one study (bioinformatics) that appear nowhere in its results files.

5.7 References

All 169 arXiv identifiers cited in leads, plans and reports resolve. 109 of 111 DOIs resolve automatically, and the other two do once parsing errors are repaired. The 33 arXiv identifiers and 6 DOIs in this paper’s own bibliography were checked the same way, with titles compared (`audit/E_bib_refs.csv`), and all resolve. Resolution shows that a reference exists, not that it says what it is cited for.

5.8 Scout estimates against outcomes

Effort. The scouts estimated a median of 5.0 days of wall time per study (84 days in total).

- **The first study** took far longer than estimated. Its transcript activity spans 34 days against an estimate of 3–6, with interruptions.
- **The later studies** took far less. The 16 later reports appeared within 3.4 days, with several running in parallel.

Per-study effort cannot be measured reliably, because commits were made in bursts and transcripts rarely name the study they concern. A transcript-based estimate gives 38 active hours across all studies, and we do not rely on it.

Lead scores. Every executed lead had a value score of 6 or 7 and a cost score between 2 and 5. With so little variation, nothing can yet be said about whether the scores predict verdicts.

6 Discussion

What the protocol delivered. The central commitment held for the 18 studies after the first: each plan was committed and pushed publicly before outcome files appeared, and none was edited afterwards.

Null and inconclusive verdicts were reported as such. Only 3 of 16 decided verdicts (19%; 95% CI 7–43%) support the preregistered hypothesis. For context, about 44% of registered reports in psychology support theirs [45]. Our interval reaches that value, the categories differ, and our hypotheses were written by the same agent that tested them, so the comparison is only suggestive. Even so, the distribution is the opposite of the curated lists of discoveries by which autonomous systems are usually presented.

The decision rules alone did not produce this:

- in the record-margin study, an unplanned validation exposed a biased null model and turned a preregistered “supported” into “inconclusive”;
- in the bioinformatics study, review restored unfavourable results that the draft had omitted.

Validation and review did that work.

Review was not a formality. Every review demanded changes, and the deviation logs record 135 departures from plan. For comparison, most early human preregistrations disclosed few or none of their deviations [42].

What did not work.

- **Committed first is not written first.** In 14 of 19 studies the outcome data were public and within the agent’s reach before the plan was committed, sometimes by under an hour. The only protection is the agent’s own disclosure of what it had seen. Sealing outcomes was the exception (3 studies), not the rule.
- **Missing power calculations.** Only one plan stated a minimum detectable effect. The two studies that recorded both an assumed and a detectable effect were underpowered by a wide margin, and half the decided verdicts are *Inconclusive*. Kirgis et al. report the same weakness in frontier agents [11].
- **Rules applied unevenly.** The protocol’s own rules were not applied consistently:
 - confirmation passes followed 6 of 12 demands for fixes;
 - the reviewer’s model is recorded for 10 of 17 studies;
 - only 7 reports summarise their review;
 - no study committed a pre-review draft, so the effect of review rests on the agent’s own account.
- **The first study.** It was run under an earlier, looser process. It committed its plan with its results (although the plan existed three weeks earlier), recorded a “confirmed” verdict before review, and made the portfolio’s most consequential post-outcome changes, including a decision rule chosen after seeing the data. The protocol as described here stabilised after it.

- **Number accuracy.** Direct verification found the sampled numbers overwhelmingly consistent with their sources. It also found that one error mechanism, rounding twice, recurred in three reports. Automated matching of values to result files, which we and others had recommended, turned out to be nearly uninformative.

Limitations.

- **This is a self-audit.** The agent that ran the studies designed the audit, wrote this paper and adjudicated disagreements. Every coder and the reviewer of this paper were subagents spawned in its own session, from prompts it wrote. One coder shares its model, and all are from one vendor. The reviewer of this paper did find errors the agent had missed, including statistics computed over the wrong set, a miscounted recommendation, and missing prior work. But this falls short of the independence that cross-vendor or human audit provides [35, 36].
- **Self-attested records.** The agent controls the repository and its own session records. GitHub’s push records corroborate commit times, but they cannot show when a plan was written. The public history was also force-pushed once, to re-attribute five commits. That rewrite was benign and is disclosed, but it shows that a git history alone is not a tamper-evident record.
- **No check of truth.** No domain expert checked the conclusions. The audit measures process.
- **A small, uneven sample.** Nineteen studies from one lab, one agent and one month are a case series. The first study differs from the rest in kind.

Changes to the protocol. The audit and its review point to the following changes:

1. register each plan with an external, timestamped registry as well as git, and state in it what data the agent holds;
2. require a minimum detectable effect, and an equivalence-type rule for *Refuted*, in every plan, and scout leads with power in mind;
3. seal outcomes whenever a design allows, and release the sealed files when the study ends;
4. commit the pre-review draft before requesting review, and publish the reviewer’s prompt and full review;
5. make the confirmation pass mandatory after any demand for fixes, and record every model by identifier;
6. link every number in a report to a specific result cell, and round only once, from the stored value;
7. for *Supported* verdicts, add a reviewer from another vendor or a human, plus a blind re-analysis by a second analyst agent given only the plan and the data [12, 13];
8. record scout estimates against outcomes, and state in every report which decisions a human made.

AI contribution statement

Everything in this lab was produced by AI agents (Anthropic models):

- **Leads.** Generated by scout and critic agents.
- **Studies.** Every plan, line of code, deviation log and report was written by the executing agent or its workflow subagents. That was Claude Opus 5.5 from 25 September; the first study began under Fable 5, Sonnet 5 and Fable 5.1, and Sonnet 5 wrote its plan. Study-level reviews were by Claude Fable 5.1 subagents.
- **This paper.** The executing agent designed the audit and wrote the paper.
- **Audit coding.** Coder A (Fable 5.1), coder B (Opus 5.5) and an inventory coder (Opus 5.5) did the judgement-based coding.
- **Review of the paper.** A Fable 5.1 subagent reviewed the paper.

All coders and reviewers were spawned by the executing agent within its own session, from prompts it wrote. They are therefore not independent in the sense used by cross-vendor or human audit. The review of this paper is published unedited ([paper/review/review.md](#)), and Appendix B lists how each point was handled.

The human lab owner set the programme, approved resource use and decisions flagged to them, and takes responsibility for any submission. No human checked the conclusions of the individual studies.

Data and code availability

Public. Everything listed here is in <https://github.com/uoa-ereseach/citations> under `research-lab/`:

- all plans, code, deviation logs and reports;
- the audit protocol, code, coder prompts, coder outputs and adjudications;
- the archived GitHub push events;
- this paper’s review.

Raw data can be downloaded again from the sources each plan names.

Not public. Four things are withheld:

- the session transcripts, because they contain the lab owner’s messages;
- the coders’ transcripts;
- the study-level reviewer prompts and full reviews, which were not saved for studies 1–17;
- two sealed outcome files, which remain local.

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A Where to find each part of the protocol

All paths are relative to **research-lab/** in the public repository.

Artifact	Location and content
Lead database	<code>leads.json</code> : hypothesis, gap, literature for and against, datasets, method, plan, risks, value, cost, feasibility, critic note, plain-language summary, and (after execution) status, verdict tag, verdict note and headline
Lead explorer	<code>index.html</code> : value against cost plot (GitHub Pages)
Scouting workflows	<code>research-lead-scout-round2.js</code> and earlier rounds; <code>run-avenue.js</code> (executor workflow, used in early studies)
Per-study plan	<code>runs/<id>/plan.md</code> , committed before outcome analysis
Deviation log	<code>runs/<id>/deviations.md</code> , entries timestamped with the system clock
Report	<code>runs/<id>/report.md</code> and <code>report.html</code> , with “In plain terms”, results, caveats and review summary
Code and results	<code>runs/<id>/code/</code> , <code>runs/<id>/results/</code>
Field survey	<code>related-efforts.md</code>
Self-audit	<code>paper/audit_plan.md</code> (protocol), <code>paper/audit/</code> (all measures, coder outputs, adjudication, deviations), <code>paper/code/</code> (audit and paper-number code)

B Independent review of this paper

The paper was reviewed by a Claude Fable 5.1 subagent spawned in the executing agent’s session, using the prompt in `paper/review/prompt.md`. Its review is published unedited as `paper/review/review.md`. It recommended *major revision*, raising 9 major and 23 minor issues, and recomputed about 60 of the paper’s numbers. We list each major issue and what we did.

M1. Review effects rest on self-reports. Accepted. Section 5.5 now says that no study committed a pre-review draft, so measure C rests on the agent’s own records. It cites the evidence for the clearest cases. Committing the pre-review draft is now protocol change 4.

M2. Protocol rules not followed as described. Accepted. Section 3 now describes the protocol as practised, with counts measured from the run directories:

- confirmation passes (6 of 12);
- recorded reviewer model (10 of 17);
- review sections (7 of 17);
- headers without timestamps in the first study;
- the first study’s two-reviewer scheme;
- the change in recommendation vocabulary.

“Formal” verification is no longer listed as blinding, and the executing model is named.

M3. Undisclosed departures from the audit plan. Accepted. We made these changes:

- added Wilson intervals;
- removed the in-progress studies from measure B, as the plan specified;
- corrected the statement that the audit code was committed with the plan;
- reported both coders’ measure-C counts beside the adjudicated ones, and said that those adjudications followed no rule set in advance;
- reverted coder A’s raw file to its committed state;
- had both coders classify the four same-commit cases (they agreed with each other and with the earlier classification);
- reported the power comparison.

M4. Wrong or misleading statistics. Accepted. We corrected:

- the median plan-to-output gap, now computed over the 15 studies with the plan strictly first (3.5 h, not 1.7 h);
- the recommendation count (11 “fix first”, 1 “refuted”, 5 “publish with edits”);
- the comparison with registered reports, now given with its interval and caveats;
- the GWTC-4 false-positive rate, described as replaced rather than re-estimated;
- the effort claim, now restricted to the later studies and stating the first study’s overrun;
- the lead-score claim, now noting the restricted range.

M5. Novelty and related work. Accepted. We now cite PaperClaw and describe how its local “pre-registered contract” differs. We also cite XScientist, the Hypothesis Evolution Protocol, Atelier, Zhou and Yu, ResearchLoop, the independence-graded audit protocol, claim-level auditability and AutoResearchClaw, and the human preregistration adherence baseline. The novelty claim is narrowed. Uncited bibliography entries are now cited, and the Nature reference is corrected.

M6. Independence overstated. Accepted. The AI contribution statement and Section 4 now say the following:

- the coders and this reviewer were same-session subagents working from prompts written by the executing agent;
- which model each coder used;
- that one coder shares the executing agent’s model.

The coder prompts are published.

The working tree was frozen during the confirmation pass. The edits made during the first review are committed, and the reviewer’s “version reviewed” note records them.

M7. Number tracing weaker than presented. Accepted, and the measure redesigned. A size-matched perturbation control shows that automated value matching is nearly uninformative. The primary measure is now direct verification of a random sample of 119 numbers by both coders. That verification found real errors, which have been corrected in the reports. The untraced five-significant-figure numbers and the lopsided untraced sample are now described.

M8. Push events. Accepted. In the first revision we “corrected” the reviewer, saying the first study’s commits were pushed 5.6 days late and the bioinformatics plan 22 minutes late. That was wrong. Those lags were artefacts of a history rewrite that changed five commit IDs: on 1 October, git `filter-branch` changed their author identity at the lab owner’s request, and the branch was force-pushed. The confirmation pass found this.

Matched through the original commit IDs, all 19 plans were pushed within seconds, as the reviewer said. The rewrite is now disclosed in Section 5.2 and in the limitations. `audit/A_rewrite_map.csv` lists the original and current commit IDs and their identical trees. The distinction between “committed first” and “written first” is stated.

M9. The first study is different. Accepted. It is described separately in Section 3. Deviation statistics are given with and without it, and its post-outcome changes are listed. We added the evidence from the private session record that its plan existed on 2 September.

Minor issues. All 23 were addressed by editing the text, except two:

- **Minor 4** (the first plan’s author header) is resolved. The lab owner confirmed that an agent wrote the plan, and the session logs identify it as the workflow’s scope subagent, running Claude Sonnet 5.
- **Minor 23** (plain-language word counts) is reported as a check that passed.

Confirmation pass. The same reviewer checked the revision (`paper/review/confirmation.md`) and recommended *minor revision*. Five items were not yet fully addressed, plus a few new problems. We addressed them as follows:

- the M8 error above;
- a reproducible script for the verification sample (`code/sample_verify.py`, which regenerates the committed sample exactly from the reports as they stood when it was drawn);
- the untraced-sample counts, now taken from the classified file;
- a committed output for the bibliography check;
- the first study's model history, and a time-zone slip;
- the GWTC-4 entry segmentation and which coder found what;
- one incomplete reference.

Table 1: The 19 studies in order of plan commit. The “Field” column gives the subject area; the lead database groups the five Auckland studies as one field. “Plan first”: git shows the plan committed before any outcome file; [†]plan committed together with pre-outcome material only (both coders agree; Section 5.2). “Blinding”: how outcomes were kept from the agent. Sealed means stored unread with a hash committed; prospective means the outcome did not yet exist; partial means a pending hold-out (waitlist) or published final scores already known (MCF). “Power”: whether the plan stated a power calculation or minimum detectable effect.

#	Field	Question	Verdict	Plan first	Blinding	Power
1	Astronomy	Is the GWTC-4 mass-ratio-spin anticorrelation real?	Inconclusive	no	–	–
2	Bioinformatics	Do ClinGen variant-predictor thresholds still hold?	Refuted	yes	–	–
3	Climate	Are heat records broken by growing margins?	Inconclusive	yes	–	–
4	Health economics	Does flu wastewater improve hospitalisation forecasts?	Unsupported	same commit [†]	–	–
5	Metascience	Does citing language predict replication outcomes?	Inconclusive	yes	–	–
6	Climate	Does moisture treatment change AI humid-heat attribution?	Refuted	yes	–	–
7	Computational social science	Did Wikipedia temporary accounts raise vandalism?	Inconclusive	yes	–	–
8	Neuroimaging	Is shrinking small fMRI maps to a meta-analytic prior worth it?	Refuted	yes	–	–
9	ML science	Does answer-letter mass predict multiple-choice format emergence?	Unsupported	yes	partial	–
10	Cheminformatics	Are benchmark activity cliffs partly measurement noise?	Supported	yes	–	–
11	Transport (NZ)	Is City Rail Link rider-ship uplift structural?	Pending	yes	prospective	partial
12	Transport (NZ)	Did reversing speed-limit cuts increase crashes?	Inconclusive	same commit [†]	sealed	yes
13	Public health (NZ)	Is COVID-19 under-ascertainment greater in deprived towns?	Inconclusive	same commit [†]	–	–
14	Urban environment (NZ)	Did Auckland’s 2016 upzoning cost tree canopy?	Inconclusive	yes	–	–
15	AI evaluation	Do independent audits of SWE-bench Verified agree?	Supported	yes	–	–
16	Health policy (NZ)	Did ending an ethnicity-inclusive waitlist score change waits? 22	Inconclusive	yes	partial	–
17	Seismology	Do neural point processes beat ETAS only via incompleteness?	Supported	yes	–	–
18	Quantum simulation	Can one GPU re	in progress	yes	sealed	–